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India can, and is following established playbooks for aligning research, industry, and government

INDIA'S DEEP TECH MOMENT

With the race for dominance in emerging technologies like AI, quantum, defence, and energy, India's RDI investment arrives at a pivotal moment

▶ THE NEWLY LAUNCHED

₹1 trillion Research, Development and Innovation (RDI) Scheme is India's bold move to channelise scientific talent into economic strength for the Viksit Bharat 2047 vision. This investment arrives at a pivotal moment.

The urgency is undeniable. India spends just 0.6% of GDP on R&D, well below the 1.8% global average and a fraction of the 3–6% spent by leaders like South Korea and Israel. Private industry accounts for only 36% of India's R&D spend, compared to ~70% in leading economies.

The RDI scheme is promising, but concessional capital alone will not drive high-risk research. We need systemic coordination.

FROM FIRST MOVER TO MISSED OPPORTUNITY

India has been on the front edge of several deep tech frontiers, yet each early lead ebbed away for some systemic fragmentation.

In the 1970s, ISRO developed its own launch vehicles which dominated global small sat markets. But its flagship PSLV rocket never evolved into a flexible, commercial launch platform like SpaceX. In the 1980s, India's Semiconductor Complex Ltd. (SCL) produced 800 nm chips,

even preceding Taiwan's TSMC. Instead of scaling up commercially, it stuck to outdated 180 nm defence chips. TSMC, founded after SCL, moved to 3 nm and today owns 35% of global foundry output. Pharma has the same story.

WHAT AILS US

These early breakthroughs stagnated because academia, industry, and government worked in silos. Innovation leaders use a "triple helix" model: universities generate ideas, companies develop them, governments enable scale-up. This transforms inventions into industries.

The dysfunction starts with concentration of public research. Consequently, a disengaged private industry provides only 5–7% of academic research funding, and Indian firms contribute a mere 0.5% of global corporate R&D, signaling an innovation-averse mindset.

SIGNALS OF A SHIFT

Underneath these challenges, there are sparks of promise. India's Patent Office received a record 90,298 applications in 2023, with 55.2% from domestic applicants, compared to just 25% in 2013. Investments in deeptech start-ups rose 78% to \$1.6 billion, compared to 23% overall, drawing interest from both specialist and generalist firms.